Assignment 5: Telecom Optimization

Course: CS101 Computational Thinking | B.Tech 1st Year

Problem Statement

Compare multiple mobile telecommunication plans based on a user's required data limit, validity days, and budget constraints. Calculate any extra charges incurred if the usage exceeds the plan's limit, and identify the single most economical option.

Algorithm

1. Define variables for user requirements (required data, days, budget).
2. Create a list containing available plans.
3. Initialize tracking variables for the minimum cost and best plan name.
4. Use a `for` loop to evaluate each plan against validity and budget constraints.
5. Calculate excess data fees if applicable.
6. Update tracking variables if a cheaper eligible plan is found.

Python Implementation

```
# User constraints
req_data_gb = 50
req_validity_days = 28
max_budget = 600

# Available Plans: (PlanName, Included_GB, Validity_Days, Base_Cost,
Extra_Cost_Per_GB)
plans = [
    ("Basic Plan", 30, 28, 299, 15),
    ("Pro Plan", 50, 28, 399, 10),
    ("Premium Plan", 100, 56, 599, 5)
]

best_plan_name = "No eligible plan found"
lowest_cost = max_budget + 1

for plan in plans:
    name, incl_data, days, base_cost, extra_fee = plan

    # Check constraints
    if days >= req_validity_days and base_cost <= max_budget:
        total_cost = base_cost

        # Calculate excess charges
        if req_data_gb > incl_data:
            excess_gb = req_data_gb - incl_data
            total_cost += excess_gb * extra_fee

        print("Evaluated " + name + " -> Total Cost: Rs.", total_cost)

        # Find minimum
        if total_cost < lowest_cost:
            lowest_cost = total_cost
            best_plan_name = name

print("\n--- Recommendation ---")
print("Most Economical Plan:", best_plan_name)
```